

B.Sc (Hons.) in CSE Part-1, 2nd Semester In-course Examination 2023

CSE 510223 Discrete Mathematics

Marks: 20

Time: 50 minuets

1. Define proposition. Verify which of the following is proposition and why? 1+5=6
- i) Samsung S21FE 5G is the fan edition of S Series.
 - ii) Hello, Students!
 - iii) Honesty is the best policy.
 - iv) BD is facing a bitter cold today.
 - v) Metro rail is a mass rapid transit system serving Dhaka.
2. Differentiate conjunction and disjunction. 3
3. Show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent by developing a series of logical equivalences. 4
4. Consider these statements, of which the first three are premises and the fourth is a valid conclusion. "All hummingbirds are richly colored." "No large birds live on honey." "Birds that do not live on honey are dull in color." "Hummingbirds are small. Transform into logical statements. 4
5. Show that the premises "It is not sunny this afternoon and it is colder than yesterday," "We will go swimming only if it is sunny," "If we do not go swimming, then we will take a canoe trip," and "If we take a canoe trip, then we will be home by sunset" lead to the conclusion "We will be home by sunset." 5
6. Define one to one and onto function with example. 3